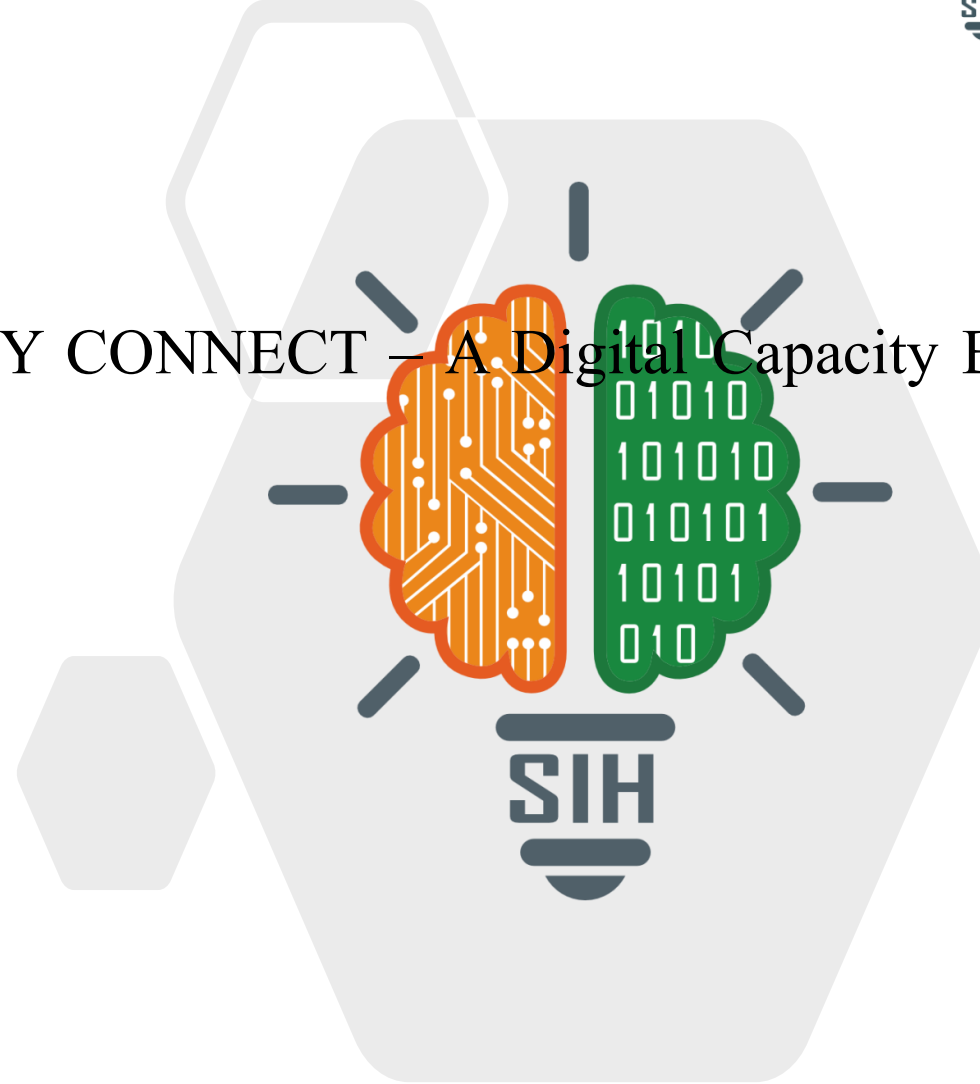


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26075
- **Problem Statement Title-** CAPACITY CONNECT – A Digital Capacity Building and Learning Management Portal.
- **Theme-** Smart Education
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal):-**TechNexus



❑ Proposed Solution

CAPACITY CONNECT is a centralized web-based Digital Capacity Building and Learning Management Portal designed to streamline organizational training, competency development, assessments, and knowledge sharing. It connects **Trainees, Trainers, and Administrators** on a single platform for efficient learning and skill development.

❑ Detailed Explanation of the Proposed Solution

Trainee

- Register and create profile
- Enroll in training programs
- Access study materials and videos
- Take MCQ assessments
- View results and provide feedback

Trainer

- Create training programs
- Upload learning resources
- Design assessments
- Monitor trainee progress
- Evaluate performance

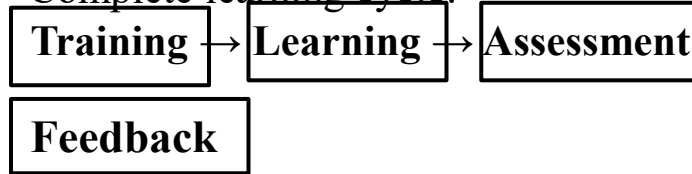
Admin

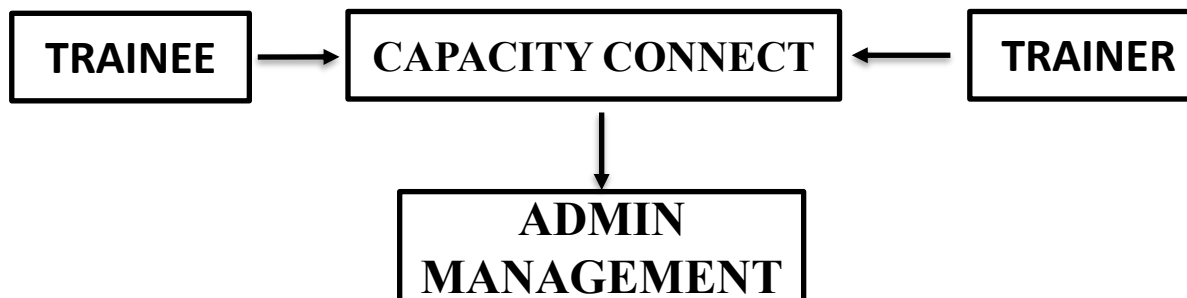
- Manage users and roles
- Approve trainer accounts
- Publish notifications
- Generate reports
- Monitor overall activities

❑ How It Addresses the Problem

- Replaces manual and scattered training systems
- Provides a centralized learning and assessment platform
- Enables competency tracking and performance monitoring
- Enables real-time communication and feedback
- Supports continuous learning and knowledge sharing

❑ Innovation and Uniqueness of the Solution

- **Role-based platform** for Trainees, Trainers, and Admins
- Complete learning cycle:

- Integrates **training, knowledge sharing, assessment, and competency development** in one platform



❑ Technologies to be Used**Frontend**

- HTML
- CSS
- JavaScript

Backend

- Python
- Flask

Database

- SQLite

Development Tools

- VS Code
- Web Browser

HTML + CSS + JavaScript**Python****SQLite****Working Web Portal****❑ Methodology / Implementation Flow****User Login / Signup****Role Selection****Dashboard****Training & Learning****Resources / Videos****MCQ Assessment****Results & Feedback****Admin Monitoring & Reports**

❑ Feasibility of the Idea

- Uses simple and easily available web technologies.
- Can be developed and deployed with low infrastructure cost.
- Modular design allows future feature expansion.
- Suitable for organizations with different training requirements.

❑ Potential Challenges & Risks

- **Data Security:** Protecting user and training data.
- **User Adoption:** Users may take time to adapt to the platform.
- **Content Management:** Maintaining updated training materials.
- **System Reliability:** Ensuring smooth performance and availability.

❑ Strategies to Overcome Challenges

- **Secure Authentication & Role-Based Access** for data protection.
- **Simple and Responsive UI** for easy adoption.
- **Admin/Trainer Controls** for managing and updating content.
- **Regular Testing & Backup** for reliability and data safety.

FEASIBILITY



Simple & Low-Cost Tech



Potential Risks



Security

Adoption

Content



- **Strategies**
- **Secure Login**
- **Simple UI**
- **Admin Controls**
- **Testing & Backup**



Reliable Platform

❑ Potential Impact on Target Audience**Trainees**

- Easy access to training and learning resources.
- Track learning progress and assessment results.
- Improve skills and competencies.

Trainers

- Easily upload and manage learning content.
- Monitor trainee participation and performance.
- Share knowledge through one platform.

Administrators

- Centralized user and training management.
- Better monitoring through reports and dashboards.
- Improved coordination of training activities.

Trainee + Trainer + Admin**CAPACITY CONNECT****Better Learning & Collaboration****Improved Skills & Organizational Growth****❑ Benefits of the Solution**

Area	Benefit
Social	Encourages knowledge sharing and collaboration
Economic	Reduces paperwork and training management cost
Educational	Improves continuous learning and skill development
Environmentl	Reduces paper usage through digital resources
Organizational	Centralizes training, assessment and feedback

❑ Reference and Research Sources

1. Smart India Hackathon

Reference for understanding the problem statement, requirements and expected solution approach.

2. Python & Flask

Reference for developing the Python-based backend and handling web application functionality.

3. HTML, CSS & JavaScript

Reference for designing the frontend and creating interactive web pages.

4. SQLite

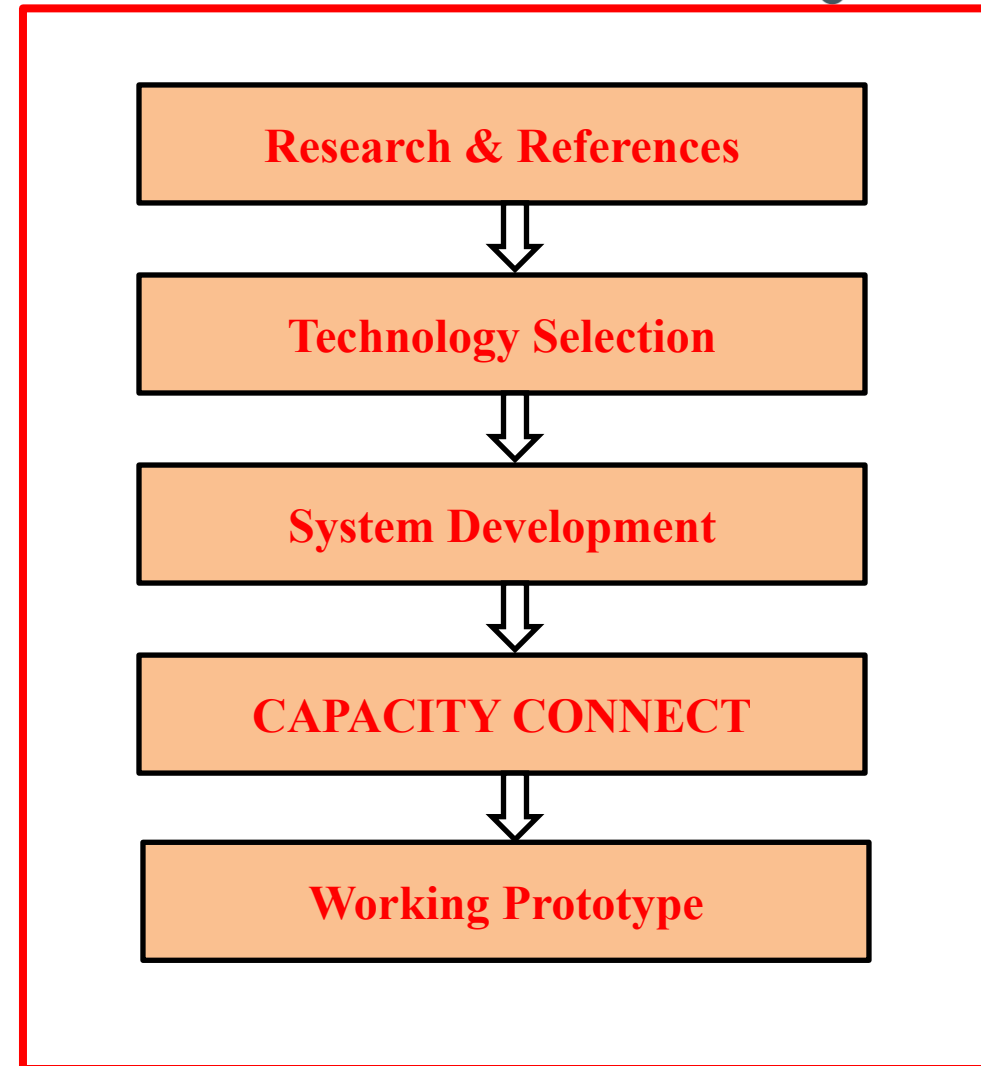
Reference for storing, managing and retrieving application data.

5. MDN Web Docs

Reference for understanding web development concepts and frontend implementation.

Reference Links

- [Smart India Hackathon](#)
- [Flask Documentation](#)
- [SQLite Documentation](#)
- [MDN Web Docs](#)
- [W3Schools](#)



These references helped us understand the required technologies and develop the Capacity Connect prototype.